5th Feb 2025

Date : _____

Module 1 : From Machine Learning to Deep Learning

• Statistics in Machine Learning

Statistics is a key component of ML with applications in:

1. Feature Engineering : Relies heavily on statistics to convert geometric features into meaningful predictors for ML algorithms.

2. Image Processing : For tasks like object detection, segmentation, statistics accurately reflect the shape and structure of objects in images.

3. Anomaly Detection : Quality control benefits from statistics by identifying deviations from norms, aiding in the detection of defects in manufacturing processes.

4. Environmental observation : Geospatial mapping leverage statistical analysis to monitor land cover patterns and ecological trends effectively.

P.T.O

- Types of Statistics:

1) Descriptive: Helps us simplify and organize big data. This makes large amounts of data easier to understand.

- Mean is calculated by summing all values present in the sample divided by total number of values present in the sample.

$$\text{Mean}(\mu) = \frac{\text{Sum of values}}{\text{No. of values}}$$

Ex. Calculate the mean of ages of a group of people: 25, 30, 27, 22, 35 and 40.

$$\therefore \text{Mean}(\mu) = \frac{25 + 30 + 27 + 22 + 35 + 40}{6}$$

$$\mu = \boxed{29.83} \text{ years}$$

- Median is defined as the middle term of the given set of data if the data is arranged in either ascending or descending order.

Ex. Find the median of ages.

22, 25, 27, 30, 35, 40

$$\text{Median} = \frac{27 + 30}{2} = \boxed{28.5} \text{ years}$$

Median for grouped data

$$\text{Median} = l + \left[\frac{\left(\frac{n}{2} - cf\right)}{f}\right] \times h$$

l = lower limit of Median class
n = no. of observations
f = frequency of Median class.
h = class size
cf = Cumulative frequency of class preceeding median class.

Ex. Find the median for given data :

Marks	No. of students	Cumulative Frequency
0 - 10	5	5
10 - 20	8	13
20 - 30	6	19
30 - 40	6	25
40 - 50	5	30

$n = 30$
$n/2 = 15$
∴ Median class = 20 - 30

$$\text{Median} = l + \left[\frac{\frac{n}{2} - cf}{f}\right] \times h$$

$$= 20 + \left[\frac{15 - 13}{6}\right] \times 10$$

$$= 20 + \frac{2}{6} \times 10$$

$$= 20 + \frac{10}{3}$$

$$= 20 + 3.33$$

$$\therefore \text{Median} = \boxed{23.33}$$

- **Mode** is the value which occurs most frequently in a set of observations.

Ex. Find mode: {6, 3, 9, 6, 6, 5, 9, 3}

Mode = $\boxed{6}$

Relationship between mean, median and mode:

$$\text{Mean} - \text{Mode} = 3[\text{Mean} - \text{Median}]$$

- **Range** is the difference between the maximum and minimum values.

- **Variance** is the average squared deviation from the mean, representing data spread.

Population variance $(\sigma^2) = \Sigma (x_i - \bar{x})^2 / n$

Sample variance $(\sigma^2) = \Sigma (x_i - \bar{x})^2 / (n-1)$

Ex. Find the population variance of data {4, 6, 8, 10}

Sol. Mean $= \frac{(4+6+8+10)}{4} = 7$

x_i	$(x_i - \bar{x})^2$
4	$(4-7)^2 = 9$
6	$(6-7)^2 = 1$
8	$(8-7)^2 = 1$
10	$(10-7)^2 = 9$

Variance $= \frac{9+1+1+9}{4} = \frac{20}{4} = \boxed{5}$

Similarly, for sample variance,

$\sigma^2 = \frac{20}{4-1} = \frac{20}{3} = \boxed{6.67}$

- **Standard deviation** is the square root of variance indicating data spread relative to the mean.

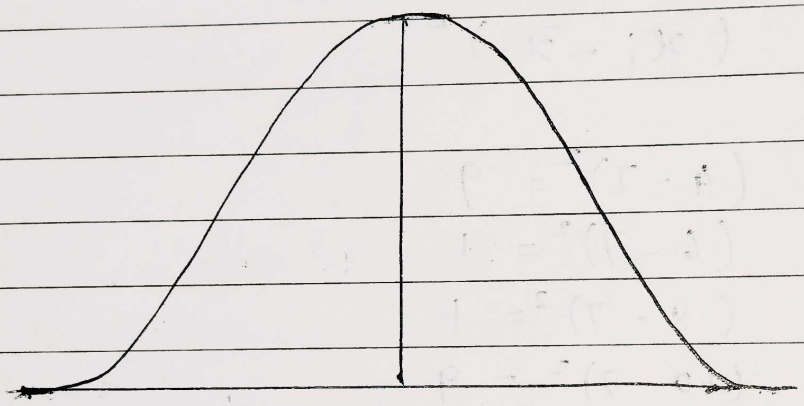
For above example,

$\sigma = \sqrt{5} = \boxed{2.236}$

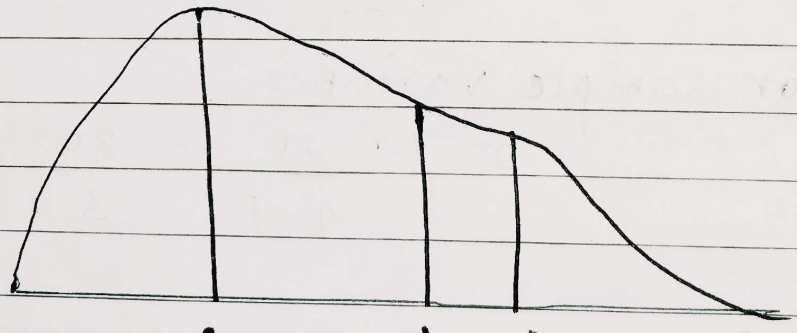
- **Interquartile Range** is the range between the first and third quartiles, measuring data spread around the media.

Measures of Shape :

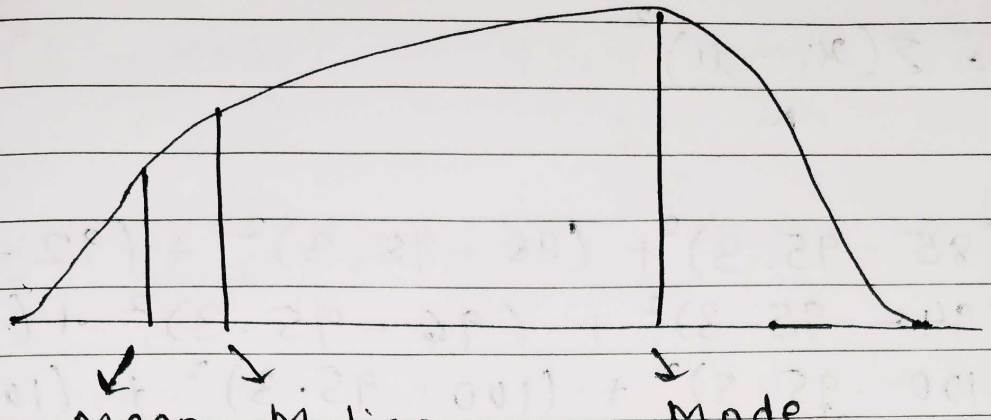
1) Skewness is a statistical measure that describes the asymmetry of the distribution of values in a dataset.



Mean = Median = Mode
No Skewness



Positive Skewness



Negative Skewness

Test of Skewness

(i) Pearson's coefficient of Skewness

Q. Calculate Pearson's skewness coefficient for a dataset of exam scores : 85, 88, 92, 94, 96, 98, 100, 100, 100, 100

Sol Step 1 : Calculate Mean

$$\bar{x} = \frac{85 + 88 + 92 + 94 + 96 + 98 + 100 + 100 + 100 + 100}{10}$$

$$= \frac{953}{10}$$

$$= 95.3$$

Step 2 : Calculate Median

$$\text{Median} = \frac{96 + 98}{2} = \frac{194}{2} = 97$$

Step 3: Calculate standard deviation.

$$\sigma^2 = \frac{\Sigma(x_i - \mu)^2}{N}$$

$$= \frac{(85-95.3)^2 + (88-95.3)^2 + (92-95.3)^2 + (94-95.3)^2 + (96-95.3)^2 + (98-95.3)^2 + (100-95.3)^2 + (100-95.3)^2 + (100-95.3)^2 + (100-95.3)^2}{10}$$

$$= \frac{268.1}{10} = 26.81$$

$$\sigma = \sqrt{26.81} \cong 5$$

Step 4: Calculate Mode.

Mode = 100

Step 5: Formulae:

(i) W.R.T Mean and Median.

$$S_k = \frac{3(\bar{x} - \text{Median})}{\sigma} = \frac{3 \times (95.3 - 97)}{5}$$

$$= -\frac{5.1}{5} = \boxed{-1.02}$$

(iii) W.R.T Mean and Mode

$$S_k = \frac{(\bar{x} - \text{Mode})}{\sigma} = \frac{95.3 - 100}{5} = \boxed{-0.94}$$

Step 6: Interpretation

S_k is negative indicating a slight negative skewness. This means the tail of skewness is slightly longer on the left side and most scores are concentrated on right side of the mean.

(ii) Bowley's Measure

Q. Same question.

Sol Step 1: Calculate median (Q_2)

$Q_2 = 97$

Step 2: Calculate Q_1

Values below mean: 85, 88, 92, 94, 96

$\therefore Q_1 = 92$

Step 3: Calculate Q_3

Values above mean: 98, 100, 100, 100, 100

$\therefore Q_3 = 100$

Step 4: Formula:

$$B = \frac{Q_1 + Q_3 - 2Q_2}{Q_3 - Q_1}$$

$$= \frac{92 + 100 - (2 \times 92)}{100 - 92}$$

$$= \frac{192 - 184}{8} = \frac{8}{8} = 1$$

$$\boxed{B = 1}$$

Step 5: Interpretation:
$B = 1$, $\therefore$ no skewness found. The data is uniformly distributed.

(iii) Kelly's Measure

Q. Same question

Sol Step 1: Find 10th percentile

$$\frac{10}{100} \times 10 = 1.$$

$\therefore$ 10th percentile = 85 $\quad P_{10} = 85$

Step 2: Find 50th percentile (Median)

$P_{50} = 97$

Step 3: Find 90th percentile

$P_{90} = 100$

Step 4: Formula:

$$Sk_L = \frac{P_{90} + 9_{10} - 2P_{50}}{P_{90} - 9_{10}}$$

$$= \frac{100 + 85 - (2 \times 97)}{100 - 85}$$

$$= \frac{185 - 194}{15} = \frac{-9}{15} = \boxed{-0.6}$$

Step 5: Interpretation.

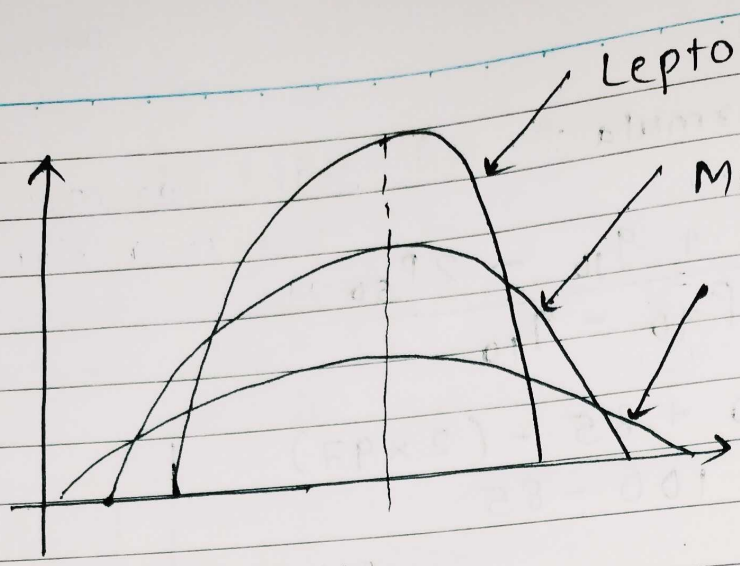
Sk_L is negative. It indicates negative skewness, meaning the distribution has a longer left tail.

2) Kurtosis

Kurtosis a characteristic of frequency distribution. It gives an idea about the shape of a frequency distribution. Basically, the measure of kurtosis is the extent to which a frequency distribution is peaked in comparison with normal curve.

Types

- LeptoKurtic
- MesoKurtic
- PlatyKurtic



Skewness	Kurtosis
1. It indicates the shape and size of variation on either side of the central value.	It indicates the frequencies of distribution at the central value.
2. The measure of differences of skewness tell us about the magnitude and direction of the asymmetry of a distribution.	It indicates the concentration of items at the central part of a distribution.
3. It indicates how far the distribution differs from the normal distribution.	It studies the divergence of the given distribution from the normal distribution

→

4.	The measure of skewness studies the extent to which deviation clusters is above or below the average.	It indicates the concentration of items.
5.	In assymmetrical distribution, the deviation below or above an average is not equal.	No such distribution takes place.

	Covariance	Correlation
1.	Covariance quantifies the interdependence of two variables.	Dividing the covariance by the sum of standard deviations of the variables.
2.	It measures the strength of the correlation between changes in multiple variables.	It standardises the covariance.
3.	Covariance is not scaled, and the units used to quantify the variables affect how much is it worth.	Correlation has a standardised measurement that has a range of -1 to +1.

4. Comparing covariances across many datasets or variables is challenging.

It enables meaningful comparisons between datasets or variables and is independent of the magnitude of the variables.

5. Co~~relation~~ variance describes ~~the~~ how much two variables vary together.

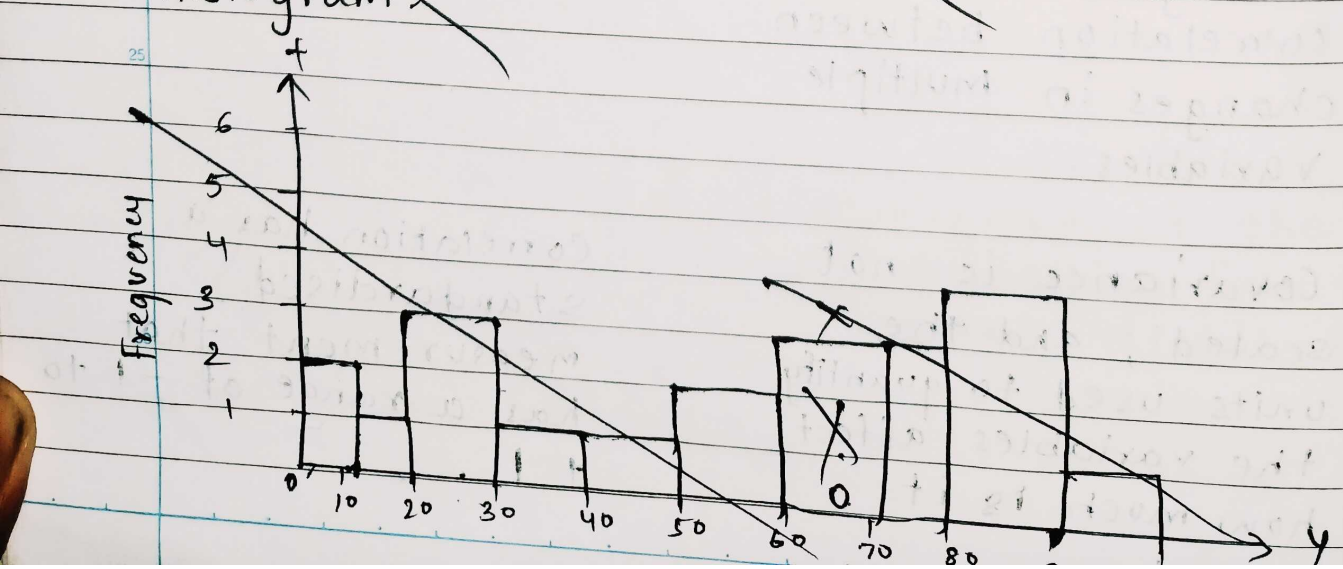
Correlation describes the intensity and direction of linear link between two variables.

- Visualization Techniques.

1) Histograms
2) Box Plots
3) Scatter Plots

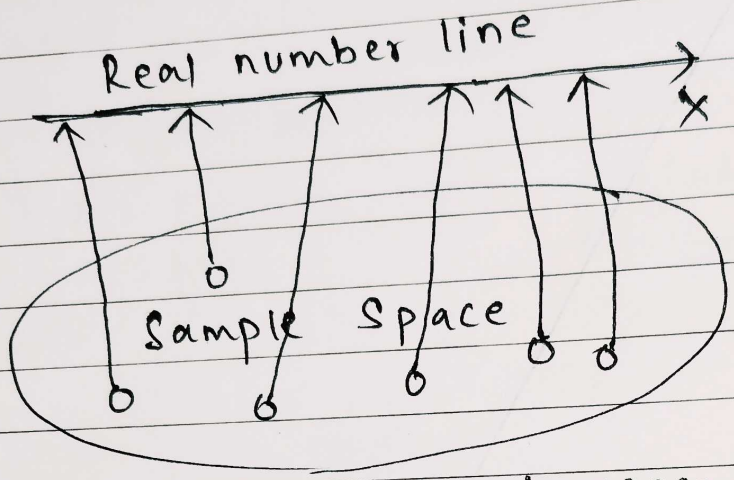
Q. 75, 81, 47, 92, 58, 2, 57, 90, 36, 80, 35, 89, 14, 78, 5, 79, 25, 80, 29, 25

Histogram:



Probability Theory

1. Random variable is a real-valued function whose domain is the sample space of the random experiment.



Events in sample space

Q. We draw two cards successively with replacement from a well-shuffled deck of 52 cards. Find the probability distribution of finding aces.

Sol There are only two aces in a deck.
$\therefore$ R.V. $x = 0$ or 1 or 2

~~P(x=0)~~ Drawing two draws with replacement means they can be considered independent experiments.

$$P(X=0) = P(\text{both cards are non-aces})$$
$$= P(\text{non-ace}) \times P(\text{non-ace})$$
$$= \frac{48}{52} \times \frac{48}{52} = \frac{144}{169}$$

(a) $P(X=6) = {}^{10}C_6 p^4 q^6$

$= {}^{10}C_6 \times \left(\frac{1}{2}\right)^4 \times \left(\frac{1}{2}\right)^6 = \frac{105}{512}$

(b) $P(X>=6)$

$= P(X=6) + P(X=7) + P(X=8) + P(X=9) + P(X=10)$

$= {}^{10}C_6 p^4 q^6 + {}^{10}C_7 p^3 q^7 + {}^{10}C_8 p^2 q^8 + {}^{10}C_9 p q^9 + {}^{10}C_{10} q^{10}$

$= \frac{193}{512}$

Poisson Distribution

Q. If 4% of the total items made by a factory are defective. Find the probability that less than 2 items are defective in the sample of 50 items.

Sol Here,

$n = 50, \quad p = \frac{4}{100} = 0.04, \quad q = (1-p) = 0.96, \quad \lambda = 2$

$PMF = \frac{\lambda^k e^{-\lambda}}{k!}$

$\therefore P(X=0) = \frac{2^0 e^{-2}}{0!} = 0.13534$

$$P(X=1) = \frac{2^1 e^{-2}}{1!} = 0.27068$$

$$\therefore P(X>2) = P(X=0) + P(X=1)$$
$$= 0.13534 + 0.27068$$
$$= \boxed{0.40602}$$

Q. If the probability of a bad reaction from medicine is 0.002, determine the chance that out of 1000 persons more than 3 will suffer a bad reaction from medicine.

Sol Here, $n = 1000$, $p = 0.002$, $\lambda = np = 2$

$$P(X>3) = 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3)]$$

$$= 1 - \left[\frac{2^0 e^{-2}}{0!} + \frac{2^1 e^{-2}}{1!} + \frac{2^2 e^{-2}}{2!} + \frac{2^3 e^{-2}}{3!}\right]$$

$$= 1 - 0.85712$$

$$= \boxed{0.1428}$$

Date :

Q. If 1% of the total screws made by a factory are defective. Find the probability that less than three screws are defective in a sample of 100.

Sol Here, $n = 100$, $p = 0.01$, $\lambda = np = 1$

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \frac{1^0 e^{-1}}{0!} + \frac{1^1 e^{-1}}{1!} + \frac{1^2 e^{-1}}{2!}$$

$$= \boxed{0.919698}$$

Q. If in an industry there is a chance that 5% of the employees will suffer from corona. What is the probability that in a group of 20 employees, more than 3 employees will suffer from corona?

Sol $n = 20$, $p = 0.05$, $\lambda = np = 1$

$$P(X > 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)]$$

$$= 1 - \left[\frac{1^0 e^{-1}}{0!} + \frac{1^1 e^{-1}}{1!} + \frac{1^2 e^{-1}}{2!} + \frac{1^3 e^{-1}}{3!}\right]$$

$$= \boxed{0.018988}$$

- visualization Techniques.

Q: Take classes (0-9), (10-19), (20-29).... and show the given data in a histogram.

36, 92, 58, 14, 70, 24, 70, 35
80, 36, 89, 2, 57, 5, 80, 35
70, 36, 80, 36, 92, 58, 2, 58
5, 82, 58, 6, 95, 81, 46, 69

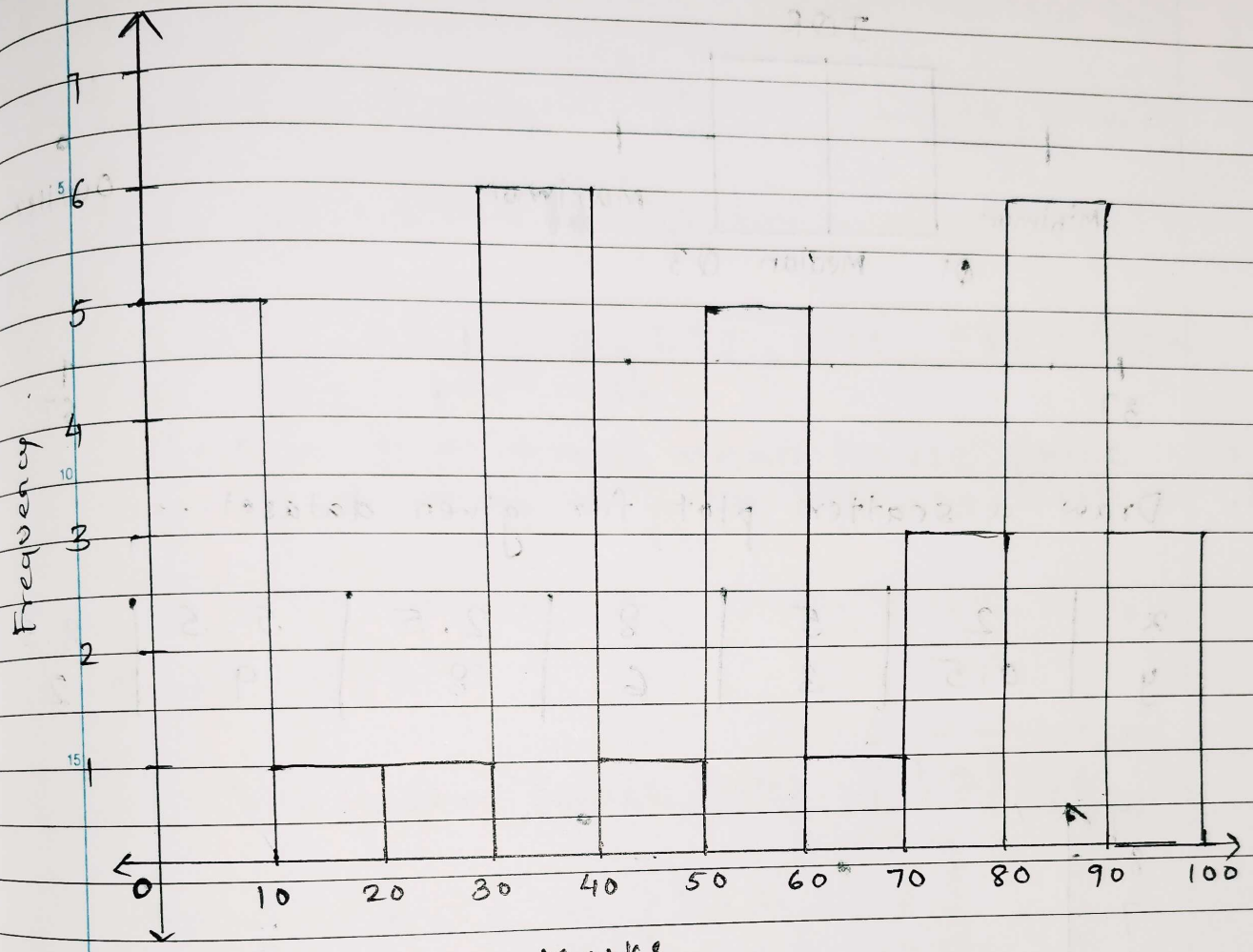
Class	Tally Marks	Frequency
0-9	𝍸	5
10-19	I	1
20-29	I	1
30-39	𝍸 I	6
40-49	I	1
50-59	𝍸	5
60-69	I	1
70-79	III	3
80-89	𝍸 I	6
90-99	III	3
		N = 32

Frequency

Q.

Sol

Date : _____



Q. Draw the Box Plot for the given data.

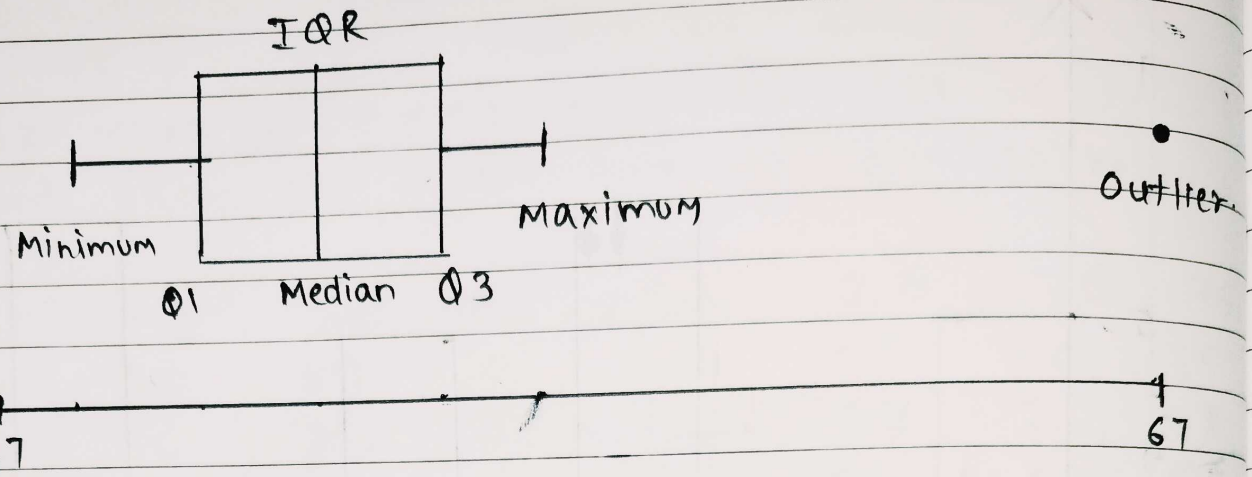
44, 47, 50, 44, 48, 50, 42, 46,
49, 67, 41, 45, 49, 43, 47, 52

Sol Ascending order : 41, 42, 43, 44, 44, 45,
46, 47, 47, 48, 49, 49, 50, 50, 52, 67

Outlier = 67

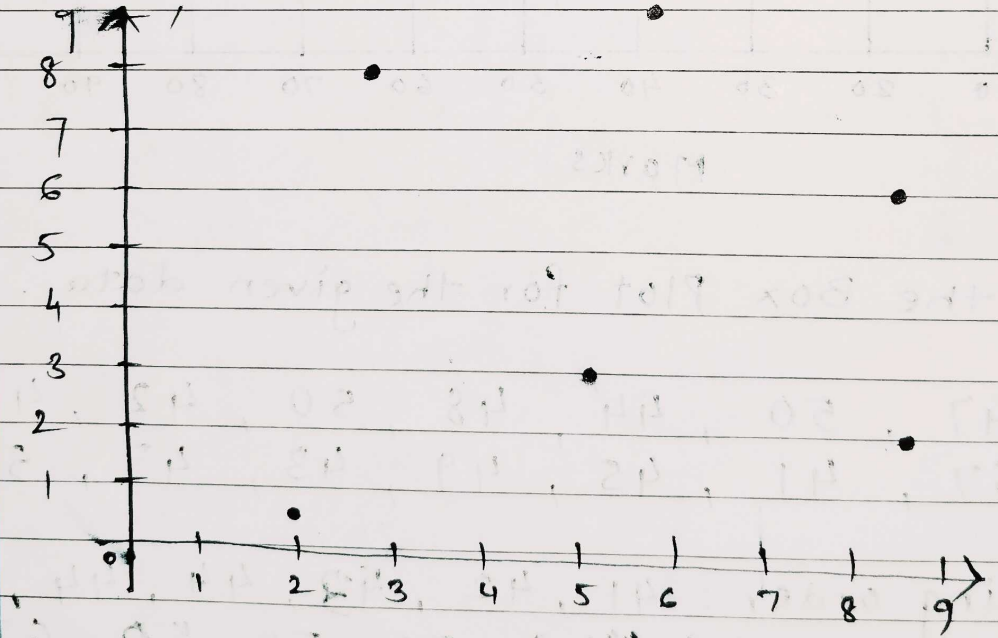
Minimum = 41
Median Q2 = 47
Q1 = 44
Q3 = 50
Maximum = 52

Date :



Q. Draw a scatter plot for given dataset.

x	2	5	8	2.5	5.5	8.5
y	0.5	3	6	8	9	2



Date :

2) Inferential Statistics

→ Estimation Techniques

- **Point Estimation (mean and variance)**

Q. Dataset : 250, 270, 290, 310, 280, 260, 275, 295, 285, 300.
Estimate the true mean house price using sample mean and point estimate by sample variance.

Sol Mean = 281.5.

∴ True mean house price = $281.5.

Variance(σ^2) = 339.17 $

P.T.O

$P(X=1)$ = P(one of the cards is ace)

= P(non ace and ace) + P(ace and non ace)

= P(non ace) × P(ace) + P(ace) × P(non ace)

$= \frac{48}{52} \times \frac{4}{52} + \frac{4}{52} \times \frac{48}{52} = \frac{24}{169}$

$P(X=2)$ = P(both cards are ace)

= P(ace) × P(ace)

$= \frac{4}{52} \times \frac{4}{52} = \frac{1}{169}$

X	0	1	2
P(X)	$\frac{144}{169}$	$\frac{24}{169}$	$\frac{1}{169}$

Q. Find the variance and mean of the number obtained on a throw of an unbiased die.

Sol Sample space = {1, 2, 3, 4, 5, 6}

$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$

X	1	2	3	4	5	6
P(X)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$E[X] = \Sigma p_i x_i$

$= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$

$$\therefore E[x] = \boxed{\frac{21}{6}}$$

$$E[x^2] = 1^2 \times \frac{1}{6} + 2^2 \times \frac{2}{6} + 3^2 \times \frac{3}{6} + 4^2 \times \frac{1}{6} + 5^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6}$$

$$= \boxed{\frac{91}{6}}$$

$$\text{variance} = E[x^2] - (E[x])^2$$

$$= \frac{91}{6} - \left(\frac{21}{6}\right)^2 = \frac{35}{12}$$

- **Binomial Distribution**

Q. When a fair coin is tossed 10 times, find the probability of getting:
(a) Exactly six heads
(b) atleast six heads

Sol $n = 10$

$p = \frac{1}{2}$ $\quad \therefore q = \frac{1}{2}$

$$P(X = x) = {}^{n}C_x \, p^{n-x} \, q^{x}$$

• Interval Estimation

Sol Step 1: Mean = $281.5

Step 2: Standard Deviation

Variance = 339.17

$\therefore s = \sqrt{339.17} = 18.42$

Step 3: Margin of Error

$$\text{Error} = \frac{s}{\sqrt{n}} = \cancel{2.6}\ \cancel{2.262}\ 5.82.$$

$t_{\alpha/2, n-1}$ for $t_{0.025, 9} = 2.262$

$\therefore$ Margin of Error $= 5.82 \times 2.262 = \boxed{13.17}$

$\therefore$ Interval:

$L = 281.5 + 13.17 = 268.33$

$H = 281.5 - 13.17 = 294.67$

Hypothesis Testing

Date :

- T-test (for $\mu = 290$)

Sol. $H_0 : \mu = 290$

$H_1 : \mu \neq 290$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$\bar{x}$ = Mean = 281.5

$s = 18.42$

$n = 10$

$\mu_0 = 29$

$\therefore t = -1.46$

$t_{0.025, 9} = 2.262$

$t < 2.262$

$\therefore$ H_0 is accepted.

• Z-test

$H_0 : \mu = 290$

$H_1 : \mu \neq 290$

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

σ = pop. std dev = 20.

$\bar{x} = 281.5$

$n = 10$

$\mu_0 = 290$.

$\therefore z = \boxed{-1.35}$

$z_{0.025} = \pm 1.96$.

$-1.35 < 1.96$

$\therefore$ $\boxed{H_0 \text{ is accepted.}}$

• **ANOVA** -

Neighbourhood A: 250, 270, 290, 310, 280
B: 260, 275, 295, 285, 300
C: 240, 255, 265, 275, 290

At 5% significance level, test if there is a statistically significant difference using ANOVA.

Sol $H_0: \mu_A = \mu_B = \mu_C$

$H_1:$ at least one μ is different.

$\overline{X}_A = 280$

$\overline{X}_B = 283$

$\overline{X}_C = 265$

$\overline{X} = \frac{280 + 283 + 265}{3} = 276$

Total sum of squares (SST)

$SST = \Sigma (X_{ij} - \overline{X})^2$

$= (280 - 276)^2 + (283 - 276)^2 + (265 - 276)^2$

$= 16 + 49 + 121 = 186$

Between group sum of squares (SSB)

$= (5 \times 16) + (5 \times 49) + (5 \times 121) = 930$

$F = \frac{MSB}{MST}$; $MSB = \frac{SSB}{k-1} = \frac{930}{2} = 465$

$MST = \frac{SST}{N-k} = \frac{186}{12} = 15.5$ $\therefore \boxed{F = 30}$

$F_{0.05,2} = 3.89$

$F > F_{0.05,2}$ $\therefore \boxed{H_0 \text{ is rejected}}$.

• Chi square test

Q.

Age Group	Prefer Product	Do not prefer Product	Total
18-30	80	70	150
31-50	60	40	100
51+	30	20	50
Total	170	130	300

at 5% LOS test whether age group & product pref are related.

Sol H_0: no association

H_1: association

$$E = \frac{\text{Row Total} \times \text{Col. Total}}{\text{Grand Total}}$$

Age group	PPO	PPE	DNPO	DNPE
18-30	80	85	70	65
31-50	60	56.67	40	43.33
51+	30	28.33	20	21.67

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \cancel{77.7556} \quad \boxed{0.92989}$$

$\chi^2_{\alpha} = 5.99$

$\chi^2 < 5.99$

$\therefore$ **Accept H_0**

Pearson correlation

Q.

X	10	12	14	15	16
Y	17	12	10	9	13

Sol

X	Y	dx	dy	dx^2	dy^2	dxdy
10	17	-3.4	4.8	11.56	23.04	-16.32
12	12	-1.4	-0.2	1.96	0.04	0.28
14	10	0.6	-2.2	0.36	4.84	-1.32
15	9	1.6	-3.2	~~10.24~~ 2.56	10.24	-5.12
16	13	2.6	0.8	6.76	0.64	2.08

$\bar{x} = 13.4$

$\bar{y} = 12.2$

$$r = \frac{\Sigma dxdy - N\bar{dx}\,\bar{dy}}{\sqrt{\left[\Sigma dx^2 - N(\bar{dx})^2\right]\left[\Sigma dy^2 - N(\bar{dy})^2\right]}}$$

$\Sigma dxdy = -20.4$ $\quad \Sigma dx^2 = 23.2$

$\bar{dx} = 0$ $\quad \Sigma dy^2 =$

$\bar{dy} = 0$

$$\therefore r = \frac{-20.4}{\sqrt{23.2 \times 38.8}} = \boxed{-0.679}$$

- **Rank correlation**

Sol

X	Y	R_x	R_y	d_i	d_i^2
10	17	1	5	−4	16
12	12	2	3	−1	1
14	10	3	2	1	1
15	9	4	1	3	9
16	13	5	4	1	1

$\Sigma d_i^2 = 28$

$$R = 1 - \left(\frac{6 \Sigma d_i^2}{N^3 - N}\right)$$

$$= 1 - \left(\frac{6 \times 28}{125 - 5}\right)$$

$$= \boxed{-0.4}$$

Date :

• Simple Linear Regression [Estimate for $x = 11$ and $y = 11$; ?]

Sol

x	y	x^2	y^2	xy
10	17	100	289	170
12	12	144	144	144
14	10	196	100	140
15	9	225	~~225~~ ~~135~~ 81	135
16	13	256	169	208

$\Sigma x = 67$ $\quad \Sigma x^2 = 921$ $\quad \Sigma xy = 797$

$\Sigma y = 61$ $\quad \Sigma y^2 = 783$

$y = A + Bx$

$\Sigma y = AN + B\Sigma x \quad \rightarrow 5A + 67B = 61$

$\Sigma xy = A\Sigma x + B\Sigma x^2 \quad \rightarrow 67A + 921B = 797$

$\therefore A = 23.9827 \; ; \quad B = -0.8793$

$\therefore y = 23.9827 - 0.8793x$

$\therefore \boxed{y = 14.3104}$

$x = A + By$

$\Sigma x = AN + B\Sigma y \quad \rightarrow 5A + 61B = 67$

$\Sigma xy = A\Sigma y + B\Sigma y^2 \quad \rightarrow 61A + 783B = 797$

$\therefore A = 19.8144 \; ; \quad B = -0.5257$

$\therefore x = 19.8144 - 0.5257y$

$x = \boxed{14.0317}$

- Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Q. A person has undertaken a job. The probabilities of completion of the job on time and without rain are 0.44 and 0.95 respectively. If the probability that it will rain is 0.45, then determine the probability that job will be completed on time.

Sol $E_1 \rightarrow$ event job will be completed on time

$E_2 \rightarrow$ event that it rains.

$P(A) = 0.45$

$P(\text{no rain}) = P(B) = 1 - 0.45 = 0.55$

$P(E_1) = 0.44$

$P(E_2) = 0.95$

$$P(E) = P(A)\,P(E_1) + P(B)\,P(E_2)$$

$$= \boxed{0.7205}$$